

TECHNIQUE MASTERING EXERCISES

A - ABSOLUTE VALUE

$$1. \text{ a. } |x - 2| = \begin{cases} x - 2 & \text{if } x \geq 2 \\ -(x - 2) & \text{if } x < 2 \end{cases} = \begin{cases} x - 2 & \text{if } x \geq 2 \\ -x + 2 & \text{if } x < 2 \end{cases}.$$

$$\text{b. } |3 + 6x| = \begin{cases} 3 + 6x & \text{if } x \geq -\frac{1}{2} \\ -(3 + 6x) & \text{if } x < -\frac{1}{2} \end{cases} = \begin{cases} 3 + 6x & \text{if } x \geq -\frac{1}{2} \\ -3 - 6x & \text{if } x < -\frac{1}{2} \end{cases}.$$

$$\text{c. } |x^2 - 1| = \begin{cases} x^2 - 1 & \text{if } x^2 - 1 \geq 0 \\ -(x^2 - 1) & \text{if } x^2 - 1 < 0 \end{cases} = \begin{cases} x^2 - 1 & \text{if } x \geq 1 \text{ or } x \leq -1 \\ -x^2 + 1 & \text{if } -1 < x < 1 \end{cases}$$

$$\text{d. } |x^4 + 4| = \begin{cases} x^4 + 4 & \text{if } x^4 + 4 \geq 0 \\ -(x^4 + 4) & \text{if } x^4 + 4 < 0 \end{cases} = x^4 + 4 \text{ for all } x \in \mathbb{R}$$

2.

$$\text{a. } x = 2 \quad \text{b. } x \neq 2 \text{ or } x \in \mathbb{R} \setminus \{2\} \quad \text{c. } x \in \emptyset \text{ (ie no solution)}$$

$$\text{d. } x \in (1, 3) \quad \text{e. } x \in \left(-\frac{7}{2}, \frac{7}{2}\right) \quad \text{f. } x \in (-\infty, -2) \cup (6, \infty)$$

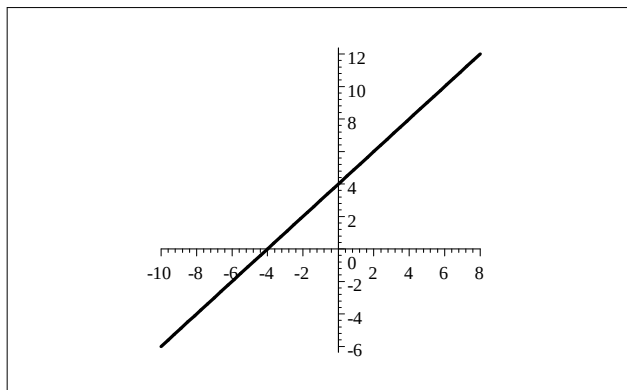
$$\text{g. } x \in [-3, 9] \quad \text{h. } x \in \left(-\infty, -\frac{11}{2}\right) \cup \left(\frac{1}{2}, \infty\right) \quad \text{i. } x \in \{-1, 2\}$$

$$\text{j. } x \in \left[-\frac{3}{7}, \frac{9}{7}\right] \quad \text{k. } x \in \emptyset \quad \text{l. } x \in (-\infty, -5) \cup \left(\frac{4}{5}, \infty\right)$$

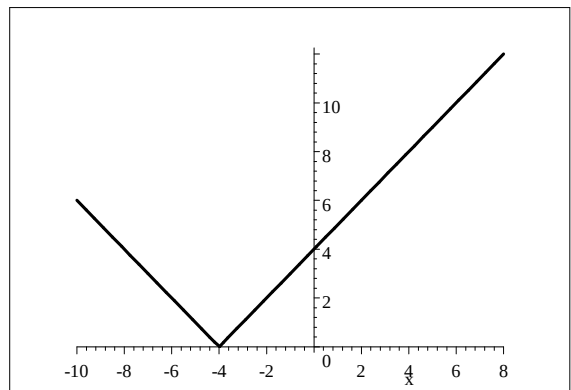
$$\text{m. } x \in \{3, -3\} \quad \text{n. } x \in \left\{-\frac{1}{5}, 1\right\} \quad \text{o. } x \in (-\infty, 0) \cup (0, 1]$$

$$\text{p. } x \in (-\infty, 1)$$

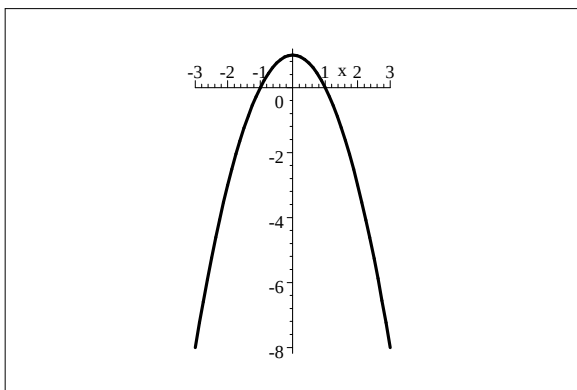
3.



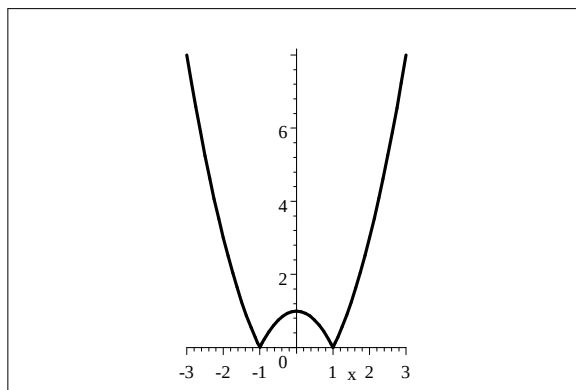
$$y = x + 4$$



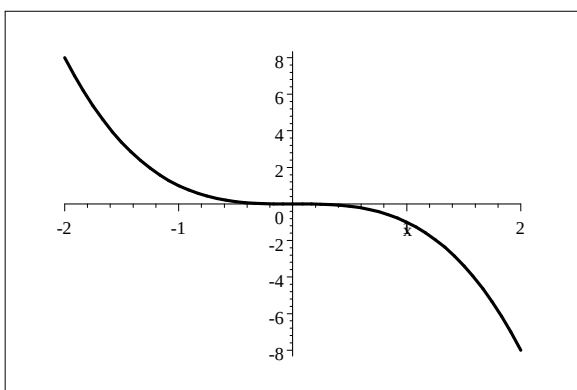
$$y = |x + 4|$$



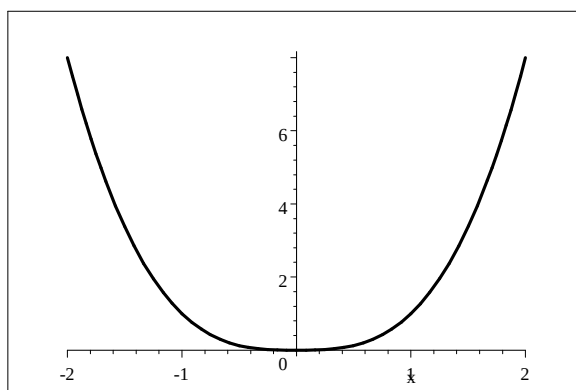
$$y = 1 - x^2$$



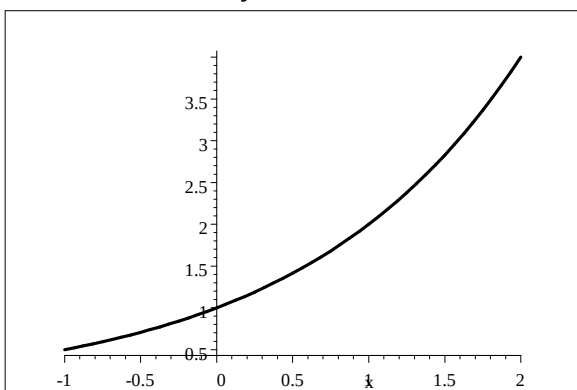
$$y = |1 - x^2|$$



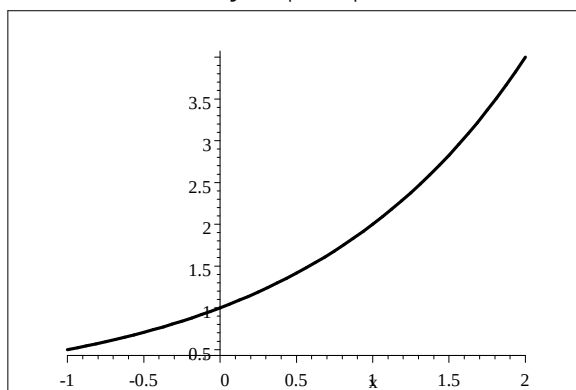
$$y = -x^3$$



$$y = |-x^3|$$



$$y = 2^x$$



$$y = |2^x|$$